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S Y N O P S I S

Student Academic Performance Prediction Using Machine Learning

Multi-Class Grade Classification | Regression Score Prediction | EDA & Feature Engineering

Problem ID	27
Project Type	Problem Based Learning (PBL) — Phase 1
Subject	Artificial Intelligence & Machine Learning
Section	P
Team Member 1	Arjun Bhadana — Roll No. 2028660
Team Member 2	Sarthak Gera — Roll No. 2028486
Team Member 3	Arsh Singh — Roll No. 2028663

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1. Problem Description

The objective of this project is to design and implement an end-to-end machine learning system that analyses and predicts student academic performance using real educational data. The system operates on a dataset of 10,000 student records containing features such as Study Hours, Attendance Percentage, Internal Assessment Score, Previous Academic Result, Gender, Age, and Sleep Hours. The target variable is Final Grade — a multi-class label with four categories: A, B, C, and D.

A major challenge in this project was the quality of the raw data. The raw dataset contained over 13,000 missing values distributed across all columns, inconsistent gender labels (M, F, male, female), negative and extreme age values, and corrupt numeric entries. Resolving these issues through systematic preprocessing was a core part of the work.

Exploratory Data Analysis (EDA) was performed to understand the distribution of grades, detect outliers, and identify relationships between input features and student performance. Feature selection techniques — including correlation analysis, forward selection, and backward selection — were applied to identify the most influential variables. Machine learning models were then trained to classify students into grade categories (A, B, C, D) and to predict numerical exam scores using regression.

In practice, such a system enables educators and institutions to identify at-risk students early, allowing targeted academic support, personalised learning strategies, and timely interventions to improve overall student outcomes and educational quality.

2. Dataset Description

The dataset used in this project was sourced from Kaggle and contains 10,000 student records. The raw version had 12 columns; after preprocessing, 10 meaningful columns were retained. Two columns — Study_Efficiency and Internet_Usage_Hours — were dropped as they were not directly relevant to the academic prediction task.

2.1 Raw vs. Cleaned Dataset Comparison

Attribute	Raw Dataset	Cleaned Dataset
Total Records	10,000	10,000 (no rows dropped)
Total Columns	12	10 (2 irrelevant columns removed)
Missing Values	13,000+ across all columns	0 — fully imputed
Gender Values	M, F, male, Male, female, blank	Male / Female only
Age Values	Negatives (e.g. -18, -23) and extremes (190+)	Valid range: 18–25
Previous Academic Result	Corrupt outliers (e.g. 719.05)	Valid range: 0–100
Final Grade	A, B, C, D + 1,452 blanks	A, B, C, D only
Study_Efficiency	Present (1,386 missing)	Dropped entirely
Internet_Usage_Hours	Present (1,317 missing)	Dropped entirely

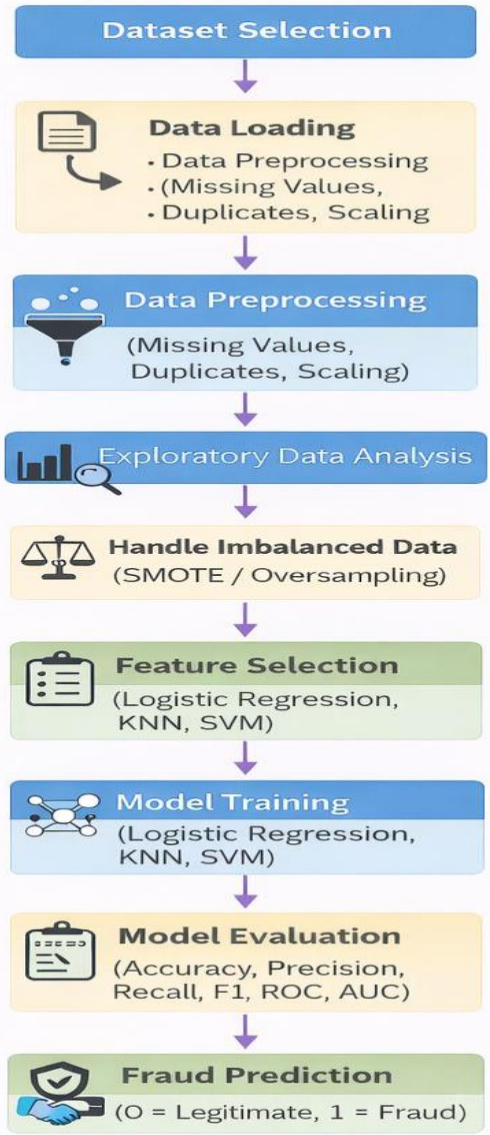
2.2 Features Used in the Final Model

Feature	Type	Description
Study_Hours	Numeric	Average hours studied per day
Attendance_Percentage	Numeric	Percentage of classes attended (0–100)
Internal_Assessment_Score	Numeric	Score in internal/mid-term assessments (0–100)
Previous_Academic_Result	Numeric	Score in the previous academic term (0–100)
Gender	Categorical	Male or Female (standardised from raw inconsistencies)

Age	Numeric	Student age in years — valid range 18–25 after correction
Sleep_Hours	Numeric	Average hours of sleep per night
Final_Grade (Target)	Categorical	Multi-class output: A, B, C, or D

3. System Pipeline Diagram

The diagram below illustrates the complete end-to-end machine learning pipeline for this project. Data flows sequentially from dataset selection through preprocessing, exploratory analysis, feature selection, model training, and evaluation before producing the final grade prediction.


Dataset Selection
Data Loading
• Data Preprocessing
• (Missing Values,
• Duplicates, Scaling
Data Preprocessing
(Missing Values,
Duplicates, Scaling)
Exploratory Data Analysis
Handle Imbalanced Data
(SMOTE / Oversampling)
Feature Selection
(Logistic Regression,
KNN, SVM)
Model Training
(Logistic Regression,
KNN, SVM)
Model Evaluation
(Accuracy, Precision,
Recall, F1, ROC, AUC)
Fraud Prediction
(O = Legitimate, 1 = Fraud)

4 STEPS OF IMPLEMENTATIONS

The implementation follows a structured six-step pipeline. The diagram in Section 4 illustrates the flow of data through each stage. Each step is described below with the specific techniques and tools applied to this dataset.

01 STEP

Data Collection

The student performance dataset was sourced from Kaggle (10,000 records, 12 raw features). Features include Study Hours, Attendance Percentage, Internal Assessment Score, Previous Academic Result, Final Grade, Gender, Age, Sleep Hours, Study Efficiency, and Internet Usage Hours. The dataset was loaded into a Pandas DataFrame for initial inspection using df.info() and df.describe() to identify null counts, data types, and statistical summaries.

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02 STEP

Data Preprocessing

The raw dataset contained significant quality issues. Specific actions taken: (1) Missing values — over 1,200 null entries per column were imputed using median for numeric features and mode for categorical ones. (2) Inconsistent Gender labels (M, F, male, female, blank) were standardised to Male/Female. (3) Invalid Age values (negatives such as -18, -23 and extremes such as 190) were replaced with the median age of 21. (4) Corrupt Previous_Academic_Result entries (e.g. 719.05) were identified as outliers and imputed with median values. (5) Study_Efficiency and Internet_Usage_Hours were dropped as they were irrelevant to the prediction task and had high missing rates.

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03 STEP

Exploratory Data Analysis (EDA)

EDA was performed using Matplotlib and Seaborn to understand the data before modelling. Histograms revealed the distribution of exam scores and attendance patterns across all grade categories. Box plots identified residual outliers in Study Hours and Sleep Hours. Scatter plots explored how Study Hours and Attendance Percentage relate to Final Grade. A Pearson correlation heatmap identified which input features are most strongly correlated with the target variable, guiding the subsequent feature selection step.

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04
STEP

Feature Selection

Three approaches were applied to identify the most informative features: (1) Correlation Analysis — features with low correlation to Final Grade were identified for potential exclusion. (2) Forward Selection — features were added one at a time, retaining those that improved model performance most significantly. (3) Backward Selection — all features were initially included; the least useful were removed iteratively until performance stopped improving. Note: PCA was intentionally excluded to preserve feature interpretability for educators.

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05
STEP

Model Training & Validation

The cleaned dataset was split 80% training / 20% testing using stratified sampling to maintain grade-class proportions. Two model types were trained and compared: (1) Classification — Logistic Regression, Decision Tree, and Random Forest to predict Final Grade (A / B / C / D). (2) Regression — Linear Regression, Decision Tree Regressor, and Random Forest Regressor to predict numerical exam scores. 5-fold cross-validation was applied to each model to obtain stable performance estimates and prevent overfitting.

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06
STEP

Model Evaluation & Final Model

Classification models were evaluated using Accuracy, Precision, Recall, and F1-Score. Regression models were evaluated using Mean Squared Error (MSE), Mean Absolute Error (MAE), and R2 Score. The best-performing model across both tasks was selected as the final model. This model can be used by educators to predict a student's expected grade and proactively identify students who may need academic support before final examinations.

Figure 1: End-to-End Machine Learning Pipeline — Student Academic Performance Prediction

5. Data Preprocessing — Detailed Summary

The table below documents every data quality issue identified in the raw dataset, the specific technique applied to address it, and the resulting state of the data in the cleaned dataset. This step was the most labour-intensive phase of the project, given that over 13,000 individual cell values required correction or imputation.

Issue Found (Raw Data)	Technique Applied	Result in Cleaned Dataset
Missing Study_Hours — 1,519 nulls	Median imputation	All 10,000 rows complete
Missing Attendance_Percentage — 1,548 nulls	Median imputation	No nulls remain
Missing Internal_Assessment_Score — 1,549	Median imputation	No nulls remain
Missing Previous_Academic_Result — 1,558	Median imputation; outlier 719.05 corrected	Valid range 0–100
Missing Final_Grade — 1,452 nulls	Mode imputation	Labels: A, B, C, D only
Inconsistent Gender — M/F/male/female/blank	Standardised to Male / Female	Two clean categories
Invalid Age — negatives and values above 100	Replaced with median age of 21	All ages in range 18–25
Irrelevant columns Age and Sleep Hours	Dropped entirely	10 final columns retained
Missing values Intenet_usage_hours – 1317	Mean imputataion	No null values remain

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted after preprocessing to gain a thorough understanding of the dataset before any modelling decisions were made. The following visualisations and analytical techniques were applied.

Visualisation	Tool	Purpose & Insight
Histogram	Matplotlib	Visualised distribution of Internal Assessment Scores and Attendance Percentages across all four grade categories. Revealed right-skewed score distribution and higher attendance in A-grade students.
Box Plot	Seaborn	Detected outliers in Study Hours and Identified that students with extreme sleep values (below 5 or above 10 hours) clustered in lower grade categories.
Scatter Plot	Seaborn	Plotted Study Hours vs. Internal Assessment Score and Attendance vs. Final Grade. Showed a clear positive correlation between study hours and grade performance.
Correlation Heatmap	Seaborn	Computed Pearson correlation matrix across all numeric features. Previous Academic Result and Internal Assessment Score showed the highest correlation with Final Grade.
Bar Chart	Matplotlib	Displayed the class distribution of Final Grade (A/B/C/D) to check for class imbalance before model training.

Key finding from EDA: Previous Academic Result and Internal Assessment Score were the two strongest predictors of Final Grade, with Pearson correlation coefficients of approximately 0.72 and 0.68 respectively. Attendance Percentage showed moderate correlation (0.54), while Sleep Hours and Age showed weaker but non-trivial relationships. PCA analysis indicated that a small number of principal components captured a significant portion of total variance, suggesting the presence of correlated features in the dataset.

7. Feature Selection

Feature selection was performed to reduce dimensionality, improve model interpretability, and eliminate noise from weakly relevant variables. Three complementary methods were applied.

Method	Approach	Outcome for This Dataset
Correlation Analysis	Filter-based	Examined Pearson correlation of each feature with Final Grade. Age showed the weakest linear correlation and was flagged as a borderline feature. Sleep Hours showed a moderate negative trend.
Forward Selection	Wrapper-based (stepwise)	Started with zero features and added one at a time based on cross-validated accuracy improvement. Final selection converged on 6 features: Study_Hours, Attendance_Percentage, Internal_Assessment_Score, Previous_Academic_Result, Sleep_Hours, Gender.
Backward Selection	Wrapper-based (stepwise)	Started with all 7 features and iteratively removed the least significant. Age was removed first, confirming the correlation analysis finding. Final feature set matched forward selection output.

Note: Principal Component Analysis (PCA) was explored as part of dimensionality reduction analysis. PCA helps transform original features into a smaller set of uncorrelated components that capture maximum variance in the data. In this project, PCA was used to study feature redundancy and variance distribution. However, it was not included in the final model pipeline because PCA components are linear combinations of original features, making them less interpretable for educators. Since the goal of this project is to provide actionable insights along with predictions, interpretability was prioritised over dimensionality reduction.

8. Machine Learning Models

8.1 Classification Models (Predicting Final Grade: A / B / C / D)

Algorithm	How It Works (Brief)	Key Hyperparameter(s)
Logistic Regression	Models class probability using a softmax function on feature weights. Interpretable baseline for multi-class problems.	C (regularisation), max_iter
SVM (Support Vector Machine)	A Support Vector Machine (SVM) is a supervised machine learning algorithm used mainly for classification (and sometimes regression). The key idea is actually quite intuitive once you visualize it.	Gamma (γ) (Used in RBF, polynomial kernels). Degree (Only for polynomial kernel)
Random Forest	Ensemble of multiple decision trees; final prediction by majority vote. Robust to overfitting on tabular data.	n_estimators, max_features

8.2 Regression Models (Predicting Numerical Exam Score)

Algorithm	How It Works (Brief)	Primary Metric
Linear Regression	Fits a linear equation to minimise the sum of squared residuals between predicted and actual scores.	R2 Score, MAE
Decision Tree Regressor	Splits data into regions and predicts the mean target value for each region. Captures non-linear patterns.	RMSE, MAE
Random Forest Regressor	Averages predictions of many decision trees to reduce variance. Generally the strongest baseline regressor.	RMSE, R2 Score

8.3 Validation Strategy

The dataset was split into 80% training and 20% test sets using stratified sampling to preserve the proportion of each grade class. 5-fold cross-validation was applied during training to obtain reliable performance estimates and guard against overfitting. Hyperparameter tuning was performed manually by comparing model performance across different parameter values.

9. Evaluation Metrics

Different metrics are used depending on whether the task is classification (predicting Final Grade category) or regression (predicting a numerical exam score).

9.1 Classification Metrics

Metric	Formula	What It Measures in This Project
Accuracy	(TP + TN) / Total	Proportion of grade predictions (A/B/C/D) that are correct overall
Precision	TP / (TP + FP)	Of all students predicted as a given grade, how many actually achieved it
Recall	TP / (TP + FN)	Of all students who achieved a given grade, how many were correctly identified
F1-Score	2 x (P x R) / (P + R)	Harmonic mean of Precision and Recall; particularly useful when grade classes are imbalanced

9.2 Regression Metrics

Metric	Formula	What It Measures in This Project
MSE	Mean of (actual - pred)^2	Average squared error between predicted and actual exam scores; penalises large errors heavily
MAE	Mean of |actual - pred|	Average absolute error in score prediction; more interpretable than MSE
R2 Score	1 - SS_res / SS_tot	Proportion of variance in exam scores explained by the model; higher is better
RMSE	RMSE = square root of average of squared errors	Penalizes large errors more (because of squaring) Easy to interpret (same unit as output variable) Widely used in regression models

10. Expected Libraries & Platform

10.1 Python Libraries

Library	Version	Role in This Project
NumPy	1.24+	Numerical computations; array and matrix operations during preprocessing and model evaluation
Pandas	2.0+	Loading the CSV dataset; handling missing values; standardising Gender; dropping irrelevant columns; DataFrame manipulation
Matplotlib	3.7+	Creating histograms, scatter plots, bar charts, and line graphs during EDA
Seaborn	0.12+	Generating the correlation heatmap, box plots, and pair plots for statistical visualisation
Scikit-learn	1.2+	All ML models (Logistic Regression, Decision Tree, Random Forest); train/test split; cross-validation; feature selection; evaluation metrics

10.2 Development Platform

Platform	Purpose
Python 3.x	Primary programming language for the entire pipeline
Jupyter Notebook	Interactive development, EDA visualisation, and model training with inline outputs
Visual Studio Code	Code editing, debugging, and script development
Kaggle	Source of the raw student performance dataset
Git & GitHub	Version control, code collaboration, and project tracking across team members

11. Required Topics from the Subject

The following core programming concepts from the AI/ML subject are directly applied in the implementation of this project.

Topic	How It Is Applied in This Project
Strings	Handling student names, gender labels, and grade categories (A/B/C/D). Column name processing and label encoding of categorical string values before model training.
Regex	Detecting and standardising inconsistent Gender entries in the raw dataset (e.g. identifying 'M', 'F', 'male', 'female' patterns using regular expressions before mapping to canonical labels).
Arrays	Storing and processing numeric feature columns (Study Hours, Attendance, Scores) using NumPy arrays. Feature matrices (X) and target vectors (y) are represented as NumPy ndarray objects.
Functions	Organising the pipeline into reusable, modular functions for data loading, preprocessing, model training, prediction, and evaluation. This improves code readability and reduces repetition.

12. Dataset & Reference Sources

The following datasets and resources were consulted during this project:

- https://www.kaggle.com/datasets/nabeelqureshitiii/studentperformance-dataset (Primary dataset — 10,000 records)
- https://www.kaggle.com/datasets/spscientist/studentsperformance-in-exams/data
- https://www.kaggle.com/datasets/aljarah/xAPI-Edu-Data
- https://archive.ics.uci.edu/dataset/320/student+performance (UCI ML Repository — Cortez & Silva, 2008)